

Harsha Kaushalya

Electrical and Information Engineering Undergraduate

✉ harshakaushalya19@gmail.com | 📞 +94-71-5906-850 | 🔗 [linkedin.com/in/HarshaKaushalya](https://www.linkedin.com/in/HarshaKaushalya)
🐙 github.com/HarshaKaushalya | 🌐 harshakaushalya.github.io/portfolio

Profile

Electrical and Information Engineering undergraduate with strong academic standing and hands-on experience in Electrical Testing/Designing, Embedded Systems and Machine Learning. Skilled in developing end-to-end solutions from hardware design to AI-powered software. Passionate about advancing knowledge through innovation and research, with a proven track record of leadership, teamwork, and excellent communication abilities.

Research Interests

- Power Systems (HV/MV/LV) Embedded Systems Digital Hardware Design (FPGA/HDL)
- AI & Software Engineering Communication Systems Internet of Things

Education

University of Ruhuna

B.Sc. in Electrical and Information Engineering

- CGPA: 3.8 | Semester 1: 3.87 | Semester 2: 3.78 | Semester 3: 3.78

Galle, Sri Lanka

Feb 2023 – Present

St. Sylvester's College

GCE Advanced Level (Physical Science Stream)

- District Rank: 111 | Z-Score: 1.778
- Physics: A, Mathematics: B, Chemistry: B

Kandy, Sri Lanka

2019 – 2021

Technical Skills

Electrical Testing: Beginner/Moderate level skills on Earth Resistance, Loop Impedance, Insulation Resistance, DC Winding Resistance and Protection Relay Testing, Thermography Surveying

Electrical Designing: Beginner level skills on Breaker Cable Coordination, ETAP, PVSyst, DiaLux

Circuit, PCB & Digital Design: Circuit Simulation, Schematic Design and PCB Design using Proteus, KiCad, EasyEDA; FPGA Design using Verilog/System Verilog in Vivado

Programming Languages: Python, JavaScript, TypeScript, SQL, C/C++, Assembly (x86)

AI/ML: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib

AI-Powered Web Development: MERN Stack, Next.js, Express.js, MySQL, RAG-based AI Systems, LLM Integration (Ollama/OpenAI), Computer Vision using YOLOv8 and OpenCV

Development Tools and Platforms: VS Code, Antigravity IDE, Cursor, Docker, MATLAB, GitHub, Figma, Arduino IDE, Cisco Packet Tracer, Google Colab, AMD Vivado

Soft Skills: Problem Solving Mindset, Effective Communication, Team Collaboration

Languages: Sinhala (Native), English (Fluent)

Projects

IoT Smart Energy Meter - Embedded Systems & Machine Learning

2025 (Completed)

- Built an ESP32-based smart energy meter measuring RMS voltage, current, frequency, power factor and kWh; streamed real-time data to a cloud dashboard with SMS/push anomaly alerts and relay-based load protection.
- Integrated manual, web-interface and voice-command load control; developed an ML load forecasting model on collected sensor data to predict consumption patterns and recommend energy-saving strategies.

Preliminary Energy Audit

2024 (Completed)

- Conducted a preliminary energy audit as part of a 26-group faculty-wide initiative, focusing on fan-load consumption across four student hostels to identify energy-saving opportunities.

- Collected and analyzed room-wise fan usage data (type, power rating, hours of operation) via Google Forms and on-site surveys, then proposed efficiency recommendations based on consumption patterns.

Custom 32-bit RISC-V Processor Core Implementation on FPGA 2026 (Ongoing)
 • Developing a 32-bit RISC-V soft-core processor using SystemVerilog, targeting a Nexys A7 Artix-7 FPGA via the AMD Vivado design flow.

Smart Laboratory Management System for DEIE Laboratories 2025 (Ongoing)
 • Developed a scalable full-stack web and mobile ecosystem featuring real-time YOLOv8 CCTV occupancy tracking and an Ollama-powered RAG AI assistant to automate laboratory administration. [GitHub]

Hardware-Accelerated Neural Network - FPGA & Deep Learning 2025 (Completed)
 • Designed a pipelined Binarized Neural Network (BNN) accelerator in Verilog on a Nexys A7-100T FPGA
 • Trained a PyTorch model for Vivado synthesis; validated via a real-time Flask/JS dashboard over UART. [GitHub]

Machine Learning Based Human Activity Recognition 2026 (Completed)
 • Built a CNN-BiLSTM model in TensorFlow/Keras to classify six human activities from 5-sensor, 15-feature accelerometer data, achieving 93.08% test accuracy and 0.9469 macro F1-score.

EXPERIENCE

Electrical Engineering Trainee <i>Amithi Power Consultants (Pvt) Ltd</i>	Colombo, Sri Lanka <i>Aug 2025 – Nov 2025</i>
• Gaining hands-on experience in electrical design and testing within a IESL Recognized professional engineering consultancy environment.	
Quality Assurance Laboratory Assistant <i>Forever Skin Naturals Pvt Ltd</i>	Pallekele, Sri Lanka <i>Jun 2024 – Aug 2024</i>
• Performed bulk testing, raw material testing, water testing, finished goods inspection, and packaging material testing in a cosmetic production environment; developed strong quality assurance and laboratory skills.	

CERTIFICATIONS

University of Ruhuna– Electrical and Information Engineering Department:

- Volcast 2.0 ETAP and Dialux Workshop Series and Certification Course
- Networking Fundamentals Workshop Series NetWiz 1.0

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

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| • Vice Chair - IEEE Ruhuna Power Electronics Chapter | 2026 – Present |
| • Team Member - Formula Student Ruhuna High Voltage Team | 2026 – Present |
| • Organizing Committee Member - Rextro Silver Jubilee Exhibition, University of Ruhuna | 2025 |
| • Editor - Telecommunication Circle, Electrical and Information Engineering Society, University of Ruhuna | 2023 – 2024 |
| • Organizing Committee Member - AeroXplore RC Plane Design Workshop and Competition, University of Ruhuna | 2024 |

VOLUNTARY WORK

- 2022 – Present: Voluntary Mathematics tutor for "Mehewara Ganitha Sawiya (Organized by University of Ruhuna)" and other organizations.
- Dec 2025: Coordinator of the AI and Computer Vision Zone of the Silver Jubilee Exhibition, University of Ruhuna.

REFERENCES

Dr. W. N. B. A. Geeth Priyankara B.Sc. Eng. Hons. (Ruhuna), M.Sc. (UiA, Norway), Ph.D. (Moratuwa) Lecturer, Dept. of Electrical and Information Engineering University of Ruhuna, Sri Lanka Phone: +94769256513 Email: geeth@eie.ruh.ac.lk	Ms. Yugani Gamlath B.Sc. Special (Wayamba) Lecturer, Dept. of Electrical and Information Engineering University of Ruhuna, Sri Lanka Phone: +94703970341 Email: yugani@eie.ruh.ac.lk
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